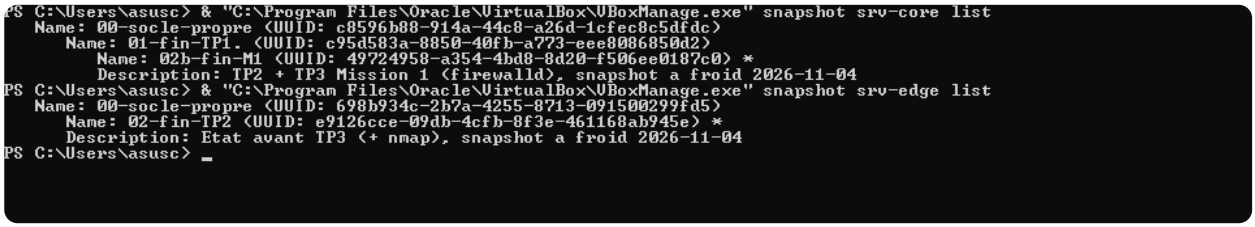
Linux Server TP3

sécuriser notre travail :

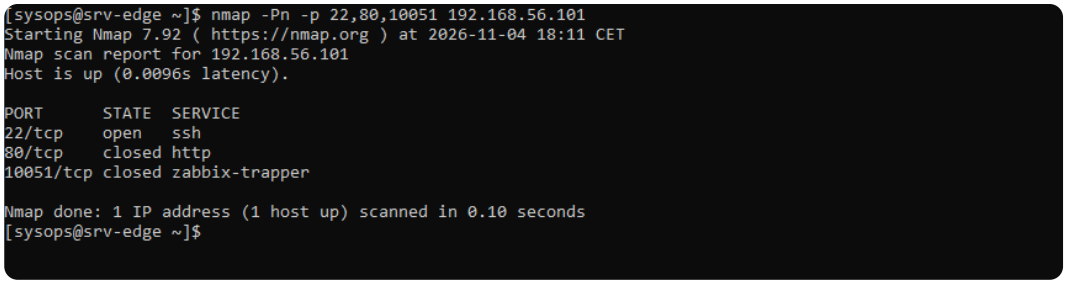
1



Mission 1 — Pare-feu

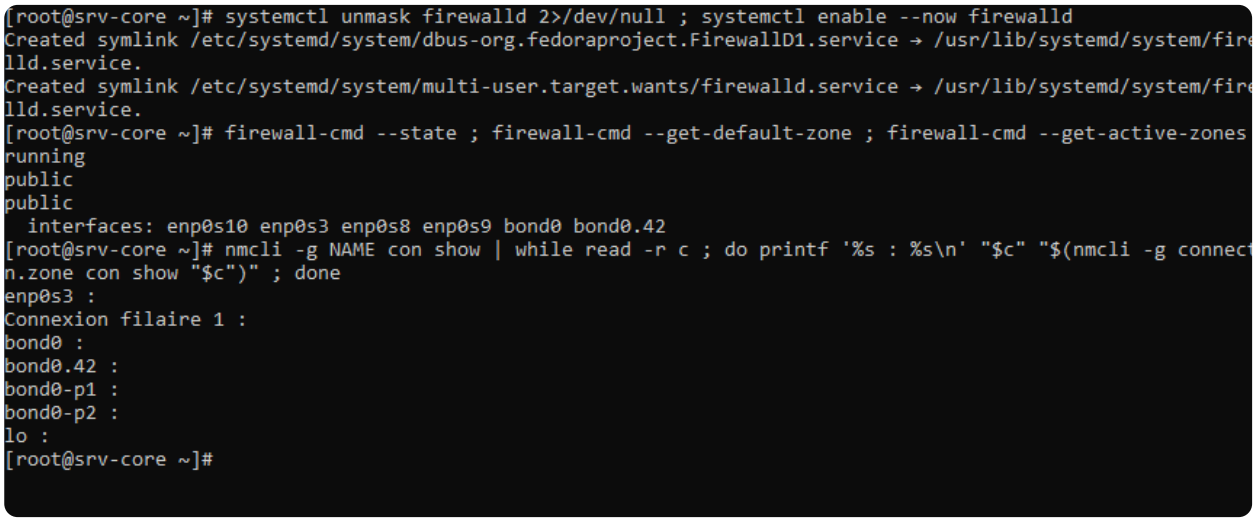
nmap avant

2



remettre firewalld en route:

3



zones des interfaces

4

```
[root@srv-core ~]# NATCON=$(nmcli -t -f NAME,DEVICE con show --active | awk -F: '$2=="enp0s3"{print $1}') ;
ho "$NATCON"
enp0s3
[root@srv-core ~]# nmcli con mod "$NATCON" connection.zone drop
[root@srv-core ~]# nmcli con mod "Connexion filaire 1" connection.zone work
[root@srv-core ~]# nmcli con mod bond0 connection.zone internal
[root@srv-core ~]# nmcli con mod bond0.42 connection.zone internal
[root@srv-core ~]# for d in enp0s3 enp0s8 bond0 bond0.42 ; do nmcli device reapply "$d" ; done
Connexion au périphérique réactivée : enp0s3)
Connexion au périphérique réactivée : enp0s8)
Connexion au périphérique réactivée : bond0)
Connexion au périphérique réactivée : bond0.42)
[root@srv-core ~]# firewall-cmd --set-default-zone=drop
Warning: ZONE_ALREADY_SET: drop
success
[root@srv-core ~]# firewall-cmd --get-default-zone ; firewall-cmd --get-active-zones
drop
drop
  interfaces: enp0s3 enp0s10
internal
  interfaces: bond0 bond0.42
public
  interfaces: enp0s9
work
  interfaces: enp0s8
[root@srv-core ~]#
```

services par zone :

5

```
[root@srv-core ~]# firewall-cmd --permanent --zone=work --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@srv-core ~]# firewall-cmd --permanent --zone=work --add-source=192.168.56.0/24
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=mdns 2>/dev/null
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=samba-client 2>/dev/null
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=cockpit 2>/dev/null
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=dhcpv6-client 2>/dev/null
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --add-port=10051/tcp
success
[root@srv-core ~]# firewall-cmd --reload
success
[root@srv-core ~]# firewall-cmd --zone=work --list-all ; firewall-cmd --zone=internal --list-all
work (active)
  target: default
  icmp-block-inversion: no
  interfaces: enp0s8
  sources: 192.168.56.0/24
  services: ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
internal (active)
  target: default
  icmp-block-inversion: no
  interfaces: bond0 bond0.42
  sources:
  services: ssh
  ports: 10051/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@srv-core ~]#
```

règles riches et ipset

6

```
[root@srv-core ~]# firewall-cmd --permanent --zone=work --remove-service=ssh
success
[root@srv-core ~]# firewall-cmd --permanent --zone=work --add-rich-rule='rule service name="ssh" log prefix="SSH-NEW " level="i
" limit value="3/m" accept limit value="6/m"'
success
[root@srv-core ~]# firewall-cmd --permanent --zone=work --add-rich-rule='rule port port="10051" protocol="tcp" reject'
success
[root@srv-core ~]# firewall-cmd --permanent --new-ipset=helios-blacklist --type=hash:ip --option=family=inet --option=hashsize=4
6 --option=maxelem=65536
success
[root@srv-core ~]# firewall-cmd --permanent --zone=work --add-rich-rule='rule source ipset="helios-blacklist" drop'
success
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --add-rich-rule='rule source ipset="helios-blacklist" drop'
success
[root@srv-core ~]# firewall-cmd --set-log-denied=unicast
success
[root@srv-core ~]# firewall-cmd --check-config && firewall-cmd --reload
success
success
[root@srv-core ~]# firewall-cmd --get-ipsets ; firewall-cmd --zone=work --list-rich-rules
helios-blacklist
rule source ipset="helios-blacklist" drop
rule service name="ssh" log prefix="SSH-NEW " level="info" limit value="3/m" accept limit value="6/m"
rule port port="10051" protocol="tcp" reject
[root@srv-core ~]# firewall-cmd --get-default-zone ; firewall-cmd --get-active-zones
drop
drop
  interfaces: enp0s3 enp0s9 enp0s10
internal
  interfaces: bond0.42 bond0
work
  interfaces: enp0s8
  sources: 192.168.56.0/24
[root@srv-core ~]#
```

ce que firewalld a écrit (nftables)

7

```
[root@srv-core ~]# nft list tables
table inet firewalld
[root@srv-core ~]# nft list chain inet firewalld filter_IN_work
table inet firewalld {
        chain filter_IN_work {
                jump filter_INPUT_POLICIES_pre
                jump filter_IN_work_pre
                jump filter_IN_work_log
                jump filter_IN_work_deny
                jump filter_IN_work_allow
                jump filter_IN_work_post
                jump filter_INPUT_POLICIES_post
                meta l4proto { icmp, ipv6-icmp } accept
                meta pkttype host log prefix "filter_IN_work_REJECT: "
                reject with icmpx admin-prohibited
        }
}
[root@srv-core ~]# nft list set inet firewalld helios-blacklist
table inet firewalld {
        set helios-blacklist {
                type ipv4_addr
                size 65536
                flags interval
        }
}
[root@srv-core ~]# iptables -S | head -5
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
[root@srv-core ~]#
```

nmap après

8

```
[sysops@srv-edge ~]$ nmap -Pn -p 22,80,10051 192.168.56.101
Starting Nmap 7.92 ( https://nmap.org ) at 2026-11-04 18:13 CET
Nmap scan report for 192.168.56.101
Host is up (0.0039s latency).

PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    filtered http
10051/tcp closed   zabbix-trapper

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
[sysops@srv-edge ~]$ nc -zv 192.168.56.101 22
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 192.168.56.101:22.
Ncat: 0 bytes sent, 0 bytes received in 0.01 seconds.
[sysops@srv-edge ~]$ nc -zv -w 2 192.168.56.101 10051 ; echo "rc attendu != 0 : $?"
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connection refused.
rc attendu != 0 : 1
[sysops@srv-edge ~]$
```

test limite 6/m

9

```
[sysops@srv-edge ~]$ for i in $(seq 1 12); do nc -zv -w1 192.168.56.101 22 ; done
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 192.168.56.101:22.
Ncat: 0 bytes sent, 0 bytes received in 0.02 seconds.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 192.168.56.101:22.
Ncat: 0 bytes sent, 0 bytes received in 0.01 seconds.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
[sysops@srv-edge ~]$
```

traces SSH-NEW

10

```
[root@srv-core ~]# journalctl -k -g 'SSH-NEW' -n 40 --no-pager | tail -8
nov. 03 17:01:14 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SRC=192.168.56.10
2 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=25816 DF PROTO=TCP SPT=48952 DPT=22 WINDOW=64240 RES=0x00 SYN U
RGP=0
nov. 03 17:01:14 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SRC=192.168.56.10
2 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=50219 DF PROTO=TCP SPT=48968 DPT=22 WINDOW=64240 RES=0x00 SYN U
RGP=0
nov. 03 17:01:16 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SRC=192.168.56.10
2 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=56750 DF PROTO=TCP SPT=48972 DPT=22 WINDOW=64240 RES=0x00 SYN U
RGP=0
nov. 03 17:01:16 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SRC=192.168.56.10
2 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=42928 DF PROTO=TCP SPT=48976 DPT=22 WINDOW=64240 RES=0x00 SYN U
RGP=0
nov. 03 17:01:43 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:0a:00:27:00:00:0f:08:00 SRC=192.168.56.1
DST=192.168.56.101 LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=38447 DF PROTO=TCP SPT=33320 DPT=22 WINDOW=65535 RES=0x00 SYN UR
GP=0
nov. 03 17:02:09 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:0a:00:27:00:00:0f:08:00 SRC=192.168.56.1
DST=192.168.56.101 LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=38488 DF PROTO=TCP SPT=46428 DPT=22 WINDOW=65535 RES=0x00 SYN UR
GP=0
nov. 03 17:02:31 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:0a:00:27:00:00:0f:08:00 SRC=192.168.56.1
DST=192.168.56.101 LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=38547 DF PROTO=TCP SPT=46433 DPT=22 WINDOW=65535 RES=0x00 SYN UR
GP=0
nov. 03 17:02:41 srv-core kernel: SSH-NEW IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:0a:00:27:00:00:0f:08:00 SRC=192.168.56.1
DST=192.168.56.101 LEN=52 TOS=0x00 PREC=0x00 TTL=128 ID=38567 DF PROTO=TCP SPT=46434 DPT=22 WINDOW=65535 RES=0x00 SYN UR
GP=0
[root@srv-core ~]#
```

11

```
[root@srv-core ~]# journalctl -k -g 'REJECT|DROP' -n 6 --no-pager
nov. 03 17:01:17 srv-core kernel: filter_IN_work_REJECT: IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SR
C=192.168.56.102 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=1605 DF PROTO=TCP SPT=46296 DPT=22 WINDOW=64240
RES=0x00 SYN URGP=0
nov. 03 17:01:18 srv-core kernel: filter_IN_work_REJECT: IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SR
C=192.168.56.102 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=26622 DF PROTO=TCP SPT=46300 DPT=22 WINDOW=64240
 RES=0x00 SYN URGP=0
nov. 03 17:01:18 srv-core kernel: filter_IN_work_REJECT: IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SR
C=192.168.56.102 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=9442 DF PROTO=TCP SPT=46312 DPT=22 WINDOW=64240
RES=0x00 SYN URGP=0
nov. 03 17:01:20 srv-core kernel: filter_IN_work_REJECT: IN=enp0s8 OUT= MAC=08:00:27:60:75:2e:08:00:27:4c:a9:43:08:00 SR
C=192.168.56.102 DST=192.168.56.101 LEN=60 TOS=0x00 PREC=0x00 TTL=64 ID=19173 DF PROTO=TCP SPT=46328 DPT=22 WINDOW=64240
 RES=0x00 SYN URGP=0
[root@srv-core ~]#
```

mission 2 — durcissement SSH

groupe ssh-admins + sysops

12

```
[root@srv-core ~]# groupadd -r ssh-admins
[root@srv-core ~]# useradd -m -c "Administrateur TP3" sysops
[root@srv-core ~]# usermod -aG ssh-admins sysops ; usermod -aG ssh-admins julien ; usermod -aG ssh-admins admin_trake
[root@srv-core ~]# getent group ssh-admins ; id sysops
ssh-admins:x:995:sysops,julien,admin_trake
uid=5104(sysops) gid=5104(sysops) groupes=5104(sysops),995(ssh-admins)
[root@srv-core ~]#
```

clé sur srv-edge

13

```
[sysops@srv-edge ~]$ ssh-keygen -t ed25519 -a 100 -C "sysops@srv-edge" -f ~/.ssh/id_ed25519 -N ''
Generating public/private ed25519 key pair.
Your identification has been saved in /home/sysops/.ssh/id_ed25519
Your public key has been saved in /home/sysops/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:oXyrbHP5S9nDwXe0vSEZgs9BbgE2+Dz7ObPZjdbz2+w sysops@srv-edge
The key's randomart image is:
+--[ED25519 256]--+
|        .+.o     |
|       .. = .    |
|        +. = .  .|
|     . . =+.o o.o|
|      o S oooo.oo|
|       . o + o..o|
|        ..+ = .. |
|     .o.o. =ooo+.|
|     .oo .o+=o +E|
+----[SHA256]-----+
[sysops@srv-edge ~]$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC11ZDI1NTE5AAAAIIpy8tQjOpaspsUwQe147qhTtU5LEiB0ZEwnOH7TWYP/ sysops@srv-edge
[sysops@srv-edge ~]$
```

14

```
[root@srv-core ~]# install -d -m 700 -o sysops -g sysops /home/sysops/.ssh
[root@srv-core ~]# echo 'ssh-ed25519 AAAAC3NzaC11ZDI1NTE5AAAAIIpy8tQjOpaspsUwQe147qhTtU5LEiB0ZEwnOH7TWYP/ sysops@srv-edg
e' > /home/sysops/.ssh/authorized_keys
[root@srv-core ~]# chown sysops:sysops /home/sysops/.ssh/authorized_keys ; chmod 600 /home/sysops/.ssh/authorized_keys ;
 restorecon -R /home/sysops/.ssh
[root@srv-core ~]# for k in /etc/ssh/ssh_host_*_key.pub ; do ssh-keygen -lf "$k" ; done
256 SHA256:4YW+zeR44gjR4IuMnDk7x/p3nymN0HmjFzGO0m/Z+ws no comment (ECDSA)
256 SHA256:9FEDKNFnN0VZNH4pM4kTE3m8zZ/lLwaEUjBdbz3KV8Y no comment (ED25519)
3072 SHA256:kivNrGORfPHjOLP3FsWYO2JJ0wYWrBKt6tutgPFqdgA no comment (RSA)
[root@srv-core ~]# ls -l /home/sysops/.ssh/authorized_keys
-rw-------. 1 sysops sysops 97  4 nov.  18:27 /home/sysops/.ssh/authorized_keys
[root@srv-core ~]#
```

test clé avant de couper le mdp

15

```
[sysops@srv-edge ~]$ ssh sysops@192.168.56.101 'hostname ; id'
srv-core
uid=5104(sysops) gid=5104(sysops) groupes=5104(sysops),995(ssh-admins) contexte=unconfined_u:unconfined_r:unconfined_t:s
0-s0:c0.c1023
```

fichier de surcharge sshd:

16

```
[root@srv-core ~]# install -m 600 -o root -g root /tmp/10-helios-hardening.conf /etc/ssh/sshd_config.d/10-helios-hardeni
ng.conf
[root@srv-core ~]# install -m 644 -o root -g root /tmp/issue.net /etc/issue.net ; restorecon -v /etc/ssh/sshd_config.d/*
 /etc/issue.net
[root@srv-core ~]# mv /etc/ssh/sshd_config.d/01-permitrootlogin.conf /root/01-permitrootlogin.conf.bak
[root@srv-core ~]# echo 'net.ipv4.ip_nonlocal_bind = 1' > /etc/sysctl.d/90-sshd-listen.conf ; sysctl -p /etc/sysctl.d/90
-sshd-listen.conf
net.ipv4.ip_nonlocal_bind = 1
[root@srv-core ~]# ls -l /etc/ssh/sshd_config.d/ ; cat /etc/issue.net
total 8
-rw-------. 1 root root 878  4 nov.  18:28 10-helios-hardening.conf
-rw-------. 1 root root 719 20 mai   02:49 50-redhat.conf
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
[root@srv-core ~]# sshd -t && echo "syntaxe OK"
syntaxe OK
[root@srv-core ~]#
```

17

```
[root@srv-core ~]# cat /etc/ssh/sshd_config.d/10-helios-hardening.conf
# --- Durcissement SSH Helios Logistique --- TP3 BTC2
Port 2242
AddressFamily inet
ListenAddress 192.168.56.101
ListenAddress 10.42.0.10

PermitRootLogin no
PubkeyAuthentication yes
PasswordAuthentication no
KbdInteractiveAuthentication no
PermitEmptyPasswords no
AuthenticationMethods publickey
MaxAuthTries 3
MaxSessions 4
MaxStartups 10:30:60
LoginGraceTime 30

AllowGroups ssh-admins

X11Forwarding no
AllowAgentForwarding no
AllowTcpForwarding no
AllowStreamLocalForwarding no
PermitTunnel no
GatewayPorts no
PermitUserEnvironment no

ClientAliveInterval 300
ClientAliveCountMax 2
TCPKeepAlive no

LogLevel VERBOSE
PrintLastLog yes
Banner /etc/issue.net
Subsystem sftp internal-sftp -l INFO -f AUTHPRIV

# Exception maîtrisée : l'equipe exploitation a besoin d'un tunnel vers le port 10051
Match Group helios-ops
    AllowTcpForwarding yes
    PermitOpen localhost:10051
[root@srv-core ~]#
```

crypto-policies

18

```
[root@srv-core ~]# update-crypto-policies --show ; cat /etc/ssh/sshd_config.d/50-redhat.conf | head -6
DEFAULT
# This system is following system-wide crypto policy. The changes to
# crypto properties (Ciphers, MACs, ...) will not have any effect in
# this or following included files. To override some configuration option,
# write it before this block or include it before this file.
# Please, see manual pages for update-crypto-policies(8) and sshd_config(5).
Include /etc/crypto-policies/back-ends/opensshserver.config
[root@srv-core ~]# update-crypto-policies --set DEFAULT:NO-SHA1
Setting system policy to DEFAULT:NO-SHA1
Note: System-wide crypto policies are applied on application start-up.
It is recommended to restart the system for the change of policies
to fully take place.
[root@srv-core ~]# update-crypto-policies --show
DEFAULT:NO-SHA1
[root@srv-core ~]# install -m 600 -o root -g root /tmp/05-helios-crypto.conf /etc/ssh/sshd_config.d/05-helios-crypto.con
f ; restorecon /etc/ssh/sshd_config.d/05-helios-crypto.conf
[root@srv-core ~]# cat /etc/ssh/sshd_config.d/05-helios-crypto.conf
HostKeyAlgorithms ssh-ed25519,rsa-sha2-512
KexAlgorithms curve25519-sha256,curve25519-sha256@libssh.org,diffie-hellman-group16-sha512
Ciphers chacha20-poly1305@openssh.com,aes256-gcm@openssh.com,aes256-ctr
MACs hmac-sha2-512-etm@openssh.com,hmac-sha2-256-etm@openssh.com
[root@srv-core ~]# sshd -t && echo "syntaxe OK"
syntaxe OK
[root@srv-core ~]#
```

port 2242 selinux + pare-feu

19

```
[root@srv-core ~]# firewall-cmd --permanent --zone=internal --remove-service=ssh
success
[root@srv-core ~]# firewall-cmd --reload
success
[root@srv-core ~]# firewall-cmd --zone=work --list-rich-rules ; firewall-cmd --zone=internal --list-all | grep -E 'servi
ces|ports'
rule port port="10051" protocol="tcp" reject
rule port port="2242" protocol="tcp" log prefix="SSH-NEW " level="info" limit value="3/m" accept limit value="6/m"
rule source ipset="helios-blacklist" drop
  services:
  ports: 10051/tcp 2242/tcp
  forward-ports:
  source-ports:
[root@srv-core ~]# ss -ltnp | grep sshd ; journalctl -u sshd -n 6 --no-pager
LISTEN 0      128    192.168.56.101:2242      0.0.0.0:*    users:(("sshd",pid=3455,fd=8))
LISTEN 0      128        10.42.0.10:2242      0.0.0.0:*    users:(("sshd",pid=3455,fd=7))
nov. 04 18:37:56 srv-core sshd-session[3551]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c fo
und at /home/admin_trake/.ssh/authorized_keys:1
nov. 04 18:37:56 srv-core sshd-session[3551]: Postponed publickey for admin_trake from 192.168.56.1 port 63436 ssh2 [pre
auth]
nov. 04 18:37:56 srv-core sshd-session[3551]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c fo
und at /home/admin_trake/.ssh/authorized_keys:1
nov. 04 18:37:56 srv-core sshd-session[3551]: Accepted publickey for admin_trake from 192.168.56.1 port 63436 ssh2: ED25
519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c
nov. 04 18:37:56 srv-core sshd-session[3551]: pam_unix(sshd:session): session opened for user admin_trake(uid=1000) by a
dmin_trake(uid=0)
nov. 04 18:37:56 srv-core sshd-session[3551]: User child is on pid 3554
[root@srv-core ~]#
```

sshd -T

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```
[root@srv-core ~]# semanage port -l | grep '^ssh_port_t'
ssh_port_t                     tcp      2242, 22
[root@srv-core ~]# sshd -T | grep -iE '^(ciphers|kexalgorithms|macs|hostkeyalgorithms)' | tr ',' '\n' | head -
20
ciphers chacha20-poly1305@openssh.com
aes256-gcm@openssh.com
aes256-ctr
macs hmac-sha2-512-etm@openssh.com
hmac-sha2-256-etm@openssh.com
kexalgorithms curve25519-sha256
curve25519-sha256@libssh.org
diffie-hellman-group16-sha512
hostkeyalgorithms ssh-ed25519
rsa-sha2-512
[root@srv-core ~]# sshd -T | grep -iE 'permitrootlogin|passwordauthentication|allowgroups|maxauthtries|^port|l
oglevel'
port 2242
maxauthtries 3
permitrootlogin no
passwordauthentication no
loglevel VERBOSE
allowgroups ssh-admins
[root@srv-core ~]#
```

les 7 tests

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```
[sysops@srv-edge ~]$ nc -zv -w2 192.168.56.101 22 ; echo "1. ancien port : echec attendu ($?)"
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: No route to host.
1. ancien port : echec attendu (1)
[sysops@srv-edge ~]$ ssh -p 2242 sysops@192.168.56.101 'hostname; id -nG'
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
srv-core
sysops ssh-admins
[sysops@srv-edge ~]$ ssh -p 2242 root@192.168.56.101 'id' ; echo "3. root : echec attendu ($?)"
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
root@192.168.56.101: Permission denied (publickey).
3. root : echec attendu (255)
[sysops@srv-edge ~]$ ssh -p 2242 -o PubkeyAuthentication=no -o PreferredAuthentications=password sysops@192.168.
56.101 ; echo "4. mot de passe : echec attendu ($?)"
Pseudo-terminal will not be allocated because stdin is not a terminal.
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
sysops@192.168.56.101: Permission denied (publickey).
4. mot de passe : echec attendu (255)
[sysops@srv-edge ~]$ ssh -p 2242 nadia@192.168.56.101 ; echo "5. hors ssh-admins : echec attendu ($?)"
Pseudo-terminal will not be allocated because stdin is not a terminal.
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
nadia@192.168.56.101: Permission denied (publickey).
5. hors ssh-admins : echec attendu (255)
[sysops@srv-edge ~]$ ssh -p 2242 -N -L 9999:localhost:10051 sysops@192.168.56.101 & sleep 2
*** Serveur Helios Logistique - acces strictement reserve ***
Toute connexion est journalisee.
[sysops@srv-edge ~]$ nc -zv -w2 127.0.0.1 9999 ; sleep 1 ; kill %1 2>/dev/null
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: Connected to 127.0.0.1:9999.
Ncat: 0 bytes sent, 0 bytes received in 0.01 seconds.
channel 2: open failed: administratively prohibited: open failed
[sysops@srv-edge ~]$ sleep 15
[sysops@srv-edge ~]$ ssh -p 2242 -o KexAlgorithms=diffie-hellman-group14-sha1 sysops@192.168.56.101 ; echo "7. K
EX SHA-1 : echec attendu ($?)"
Pseudo-terminal will not be allocated because stdin is not a terminal.
Unable to negotiate with 192.168.56.101 port 2242: no matching key exchange method found. Their offer: curve2551
9-sha256,curve25519-sha256@libssh.org,diffie-hellman-group16-sha512,ext-info-s,kex-strict-s-v00@openssh.com
7. KEX SHA-1 : echec attendu (255)
[sysops@srv-edge ~]$
```

dans le journal

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```
[root@srv-core ~]# journalctl -u sshd --since '-10min' -o short-iso --no-pager | grep -v admin_trake | tail -26
2026-11-04T18:41:44+0100 srv-core sshd-session[3970]: Connection from 192.168.56.102 port 35406 on 192.168.56.101 port 2242 rdomai
n ""
2026-11-04T18:41:44+0100 srv-core sshd-session[3970]: User root from 192.168.56.102 not allowed because none of user's groups are
listed in AllowGroups
2026-11-04T18:41:44+0100 srv-core sshd-session[3970]: Connection closed by invalid user root 192.168.56.102 port 35406 [preauth]
2026-11-04T18:41:44+0100 srv-core sshd[3455]: srclimit_penalise: ipv4: new 192.168.56.102/32 deferred penalty of 1 seconds for pen
alty: connections without attempting authentication
2026-11-04T18:41:44+0100 srv-core sshd-session[3972]: main: SHA1 in signatures is disabled for RSA keys
2026-11-04T18:41:44+0100 srv-core sshd-session[3972]: Connection from 192.168.56.102 port 35416 on 192.168.56.101 port 2242 rdomai
n ""
2026-11-04T18:41:44+0100 srv-core sshd-session[3972]: Connection closed by authenticating user sysops 192.168.56.102 port 35416 [p
reauth]
2026-11-04T18:41:44+0100 srv-core sshd-session[3974]: main: SHA1 in signatures is disabled for RSA keys
2026-11-04T18:41:44+0100 srv-core sshd-session[3974]: Connection from 192.168.56.102 port 35420 on 192.168.56.101 port 2242 rdomai
n ""
2026-11-04T18:41:44+0100 srv-core sshd-session[3974]: User nadia from 192.168.56.102 not allowed because none of user's groups are
 listed in AllowGroups
2026-11-04T18:41:44+0100 srv-core sshd-session[3974]: Connection closed by invalid user nadia 192.168.56.102 port 35420 [preauth]
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: main: SHA1 in signatures is disabled for RSA keys
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: Connection from 192.168.56.102 port 35434 on 192.168.56.101 port 2242 rdomai
n ""
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: Accepted key ED25519 SHA256:oXyrbHP5S9nDwXe0vSEZgs9BbgE2+Dz7ObPZjdbz2+w foun
d at /home/sysops/.ssh/authorized_keys:1
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: Postponed publickey for sysops from 192.168.56.102 port 35434 ssh2 [preauth]

2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: Accepted key ED25519 SHA256:oXyrbHP5S9nDwXe0vSEZgs9BbgE2+Dz7ObPZjdbz2+w foun
d at /home/sysops/.ssh/authorized_keys:1
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: Accepted publickey for sysops from 192.168.56.102 port 35434 ssh2: ED25519 S
HA256:oXyrbHP5S9nDwXe0vSEZgs9BbgE2+Dz7ObPZjdbz2+w
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: pam_unix(sshd:session): session opened for user sysops(uid=5104) by sysops(u
id=0)
2026-11-04T18:41:44+0100 srv-core sshd-session[3976]: User child is on pid 3979
2026-11-04T18:42:02+0100 srv-core sshd-session[3997]: main: SHA1 in signatures is disabled for RSA keys
2026-11-04T18:42:02+0100 srv-core sshd-session[3997]: Connection from 192.168.56.102 port 44030 on 192.168.56.101 port 2242 rdomai
n ""
2026-11-04T18:42:02+0100 srv-core sshd-session[3997]: Unable to negotiate with 192.168.56.102 port 44030: no matching key exchange
 method found. Their offer: diffie-hellman-group14-sha1,ext-info-c,kex-strict-c-v00@openssh.com [preauth]
2026-11-04T18:42:02+0100 srv-core sshd[3455]: srclimit_penalise: ipv4: new 192.168.56.102/32 deferred penalty of 1 seconds for pen
alty: connections without attempting authentication
2026-11-04T18:42:33+0100 srv-core sshd-session[4023]: main: SHA1 in signatures is disabled for RSA keys
2026-11-04T18:42:33+0100 srv-core sshd-session[4023]: Connection from 192.168.56.1 port 32483 on 192.168.56.101 port 2242 rdomain
""
2026-11-04T18:42:33+0100 srv-core sshd-session[4023]: User child is on pid 4026
[root@srv-core ~]#
```

Mission 3 — journaux

journald persistant :

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```
[root@srv-core ~]# mkdir -p /etc/systemd/journald.conf.d
[root@srv-core ~]# install -m 644 -o root -g root /tmp/50-helios.conf /etc/systemd/journald.conf.d/50-helios.conf ; cat
/etc/systemd/journald.conf.d/50-helios.conf
[Journal]
Storage=persistent
Compress=yes
SystemMaxUse=500M
SystemKeepFree=1G
SystemMaxFileSize=50M
MaxRetentionSec=1month
MaxFileSec=1day
RateLimitIntervalSec=30s
RateLimitBurst=10000
ForwardToSyslog=no
Seal=yes
[root@srv-core ~]# systemctl restart systemd-journald ; journalctl --flush
[root@srv-core ~]# ls -ld /var/log/journal/$(cat /etc/machine-id)
drwxr-xr-x. 2 root root 28  4 nov.  18:49 /var/log/journal/1fe48b93563441e49c74457e5ae30ea2
[root@srv-core ~]# journalctl --disk-usage ; journalctl --verify 2>&1 | tail -5
Archived and active journals take up 8.0M in the file system.
PASS: /var/log/journal/1fe48b93563441e49c74457e5ae30ea2/system.journal
[root@srv-core ~]#
```

après reboot

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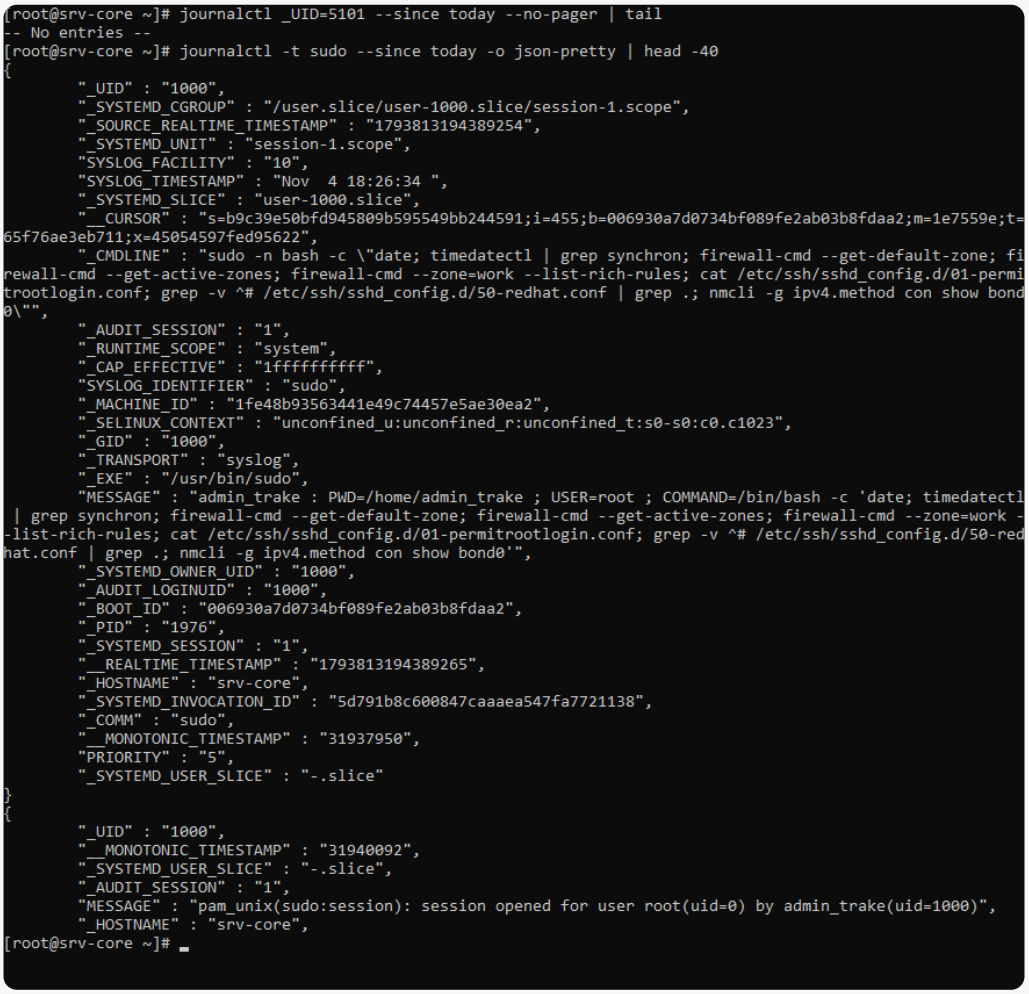
```
[root@srv-core ~]# journalctl --list-boots
IDX BOOT ID                          FIRST ENTRY                 LAST ENTRY
 -1 006930a7d0734bf089fe2ab03b8fdaa2 Wed 2026-11-04 18:26:04 CET Wed 2026-11-04 18:50:50 CET
  0 bab8b16e80f4488c889c0027fcbc118d Wed 2026-11-04 18:50:56 CET Wed 2026-11-04 18:52:09 CET
[root@srv-core ~]# ss -ltnp | grep sshd ; firewall-cmd --get-default-zone ; firewall-cmd --get-active-zones
LISTEN 0      128        10.42.0.10:2242      0.0.0.0:*    users:(("sshd",pid=856,fd=7))
LISTEN 0      128    192.168.56.101:2242      0.0.0.0:*    users:(("sshd",pid=856,fd=8))
drop
drop
  interfaces: enp0s3 enp0s10 enp0s9
internal
  interfaces: bond0 bond0.42
work
  interfaces: enp0s8
  sources: 192.168.56.0/24
[root@srv-core ~]#
```

requêtes d'investigation

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```
[root@srv-core ~]# journalctl -u sshd.service --since '-2h' -p warning -o short-iso --no-pager | tail -8
2026-11-04T18:35:33+0100 srv-core sshd[3358]: fatal: Cannot bind any address.
2026-11-04T18:35:33+0100 srv-core systemd[1]: sshd.service: Failed with result 'exit-code'.
2026-11-04T18:35:33+0100 srv-core systemd[1]: Failed to start OpenSSH server daemon.
2026-11-04T18:36:15+0100 srv-core sshd[3405]: error: Bind to port 2242 on 10.42.0.10 failed: Permission denied.
2026-11-04T18:36:15+0100 srv-core sshd[3405]: error: Bind to port 2242 on 192.168.56.101 failed: Permission denied.
2026-11-04T18:36:15+0100 srv-core sshd[3405]: fatal: Cannot bind any address.
2026-11-04T18:36:15+0100 srv-core systemd[1]: sshd.service: Failed with result 'exit-code'.
2026-11-04T18:36:15+0100 srv-core systemd[1]: Failed to start OpenSSH server daemon.
[root@srv-core ~]# journalctl _SYSTEMD_UNIT=sshd.service -g 'Invalid user|Failed|Accepted' --no-pager | tail -8
nov. 04 18:50:48 srv-core sshd-session[4582]: Accepted publickey for admin_trake from 192.168.56.1 port 15582 ssh2: ED25519 SHA256
:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c
-- Boot bab8b16e80f4488c889c0027fcbc118d --
nov. 04 18:52:09 srv-core sshd-session[2000]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c found at /ho
me/admin_trake/.ssh/authorized_keys:1
nov. 04 18:52:09 srv-core sshd-session[2000]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c found at /ho
me/admin_trake/.ssh/authorized_keys:1
nov. 04 18:52:09 srv-core sshd-session[2000]: Accepted publickey for admin_trake from 192.168.56.1 port 16705 ssh2: ED25519 SHA256
:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c
nov. 04 18:52:35 srv-core sshd-session[2069]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c found at /ho
me/admin_trake/.ssh/authorized_keys:1
nov. 04 18:52:35 srv-core sshd-session[2069]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c found at /ho
me/admin_trake/.ssh/authorized_keys:1
nov. 04 18:52:35 srv-core sshd-session[2069]: Accepted publickey for admin_trake from 192.168.56.1 port 48479 ssh2: ED25519 SHA256
:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c
[root@srv-core ~]# journalctl --facility=authpriv --since today --no-pager | tail -8
nov. 04 18:52:35 srv-core sshd-session[2069]: Accepted key ED25519 SHA256:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c found at /ho
me/admin_trake/.ssh/authorized_keys:1
nov. 04 18:52:35 srv-core sshd-session[2069]: Accepted publickey for admin_trake from 192.168.56.1 port 48479 ssh2: ED25519 SHA256
:ZZbF8E/tGAiOPBcy0eXkMQfdgBufeZD04Sh7G+/gf6c
nov. 04 18:52:35 srv-core systemd[2073]: pam_unix(systemd-user:session): session opened for user admin_trake(uid=1000) by admin_tr
ake(uid=0)
nov. 04 18:52:35 srv-core sshd-session[2069]: pam_unix(sshd:session): session opened for user admin_trake(uid=1000) by admin_trake
(uid=0)
nov. 04 18:52:35 srv-core sshd-session[2069]: User child is on pid 2080
nov. 04 18:52:35 srv-core sshd-session[2080]: Starting session: command for admin_trake from 192.168.56.1 port 48479 id 0
nov. 04 18:52:35 srv-core sudo[2109]: admin_trake : PWD=/home/admin_trake ; USER=root ; COMMAND=/bin/bash /tmp/.cap.sh
nov. 04 18:52:35 srv-core sudo[2109]: pam_unix(sudo:session): session opened for user root(uid=0) by admin_trake(uid=1000)
[root@srv-core ~]#
```

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[root@srv-core ~]# journalctl _UID=5101 --since today --no-pager | tail
-- No entries --
[root@srv-core ~]# journalctl -t sudo --since today -o json-pretty | head -40
{
"_UID" : "1000",
"_SYSTEMD_CGROUP" : "/user.slice/user-1000.slice/session-1.scope",
"_SOURCE_REALTIME_TIMESTAMP" : "1793813194389254",
"_SYSTEMD_UNIT" : "session-1.scope",
"SYSLOG_FACILITY" : "10",
"SYSLOG_TIMESTAMP" : "Nov 4 18:26:34 ",
"_SYSTEMD_SLICE" : "user-1000.slice",
"__CURSOR" : "s=b9c39e50bfd945809b595549bb244591;i=455;b=006930a7d0734bf089fe2ab03b8fdaa2;m=1e7559e;t=
65f76ae3eb711;x=45054597fed95622",
"_CMDLINE" : "sudo -n bash -c \"date; timedatectl | grep synchron; firewall-cmd --get-default-zone; fi
rewall-cmd --get-active-zones; firewall-cmd --zone=work --list-rich-rules; cat /etc/ssh/sshd_config.d/01-permi
trootlogin.conf; grep -v ^# /etc/ssh/sshd_config.d/50-redhat.conf | grep .; nmcli -g ipv4.method con show bond
0\"",
"_AUDIT_SESSION" : "1",
"_RUNTIME_SCOPE" : "system",
"_CAP_EFFECTIVE" : "1ffffffffff",
"SYSLOG_IDENTIFIER" : "sudo",
"_MACHINE_ID" : "1fe48b93563441e49c74457e5ae30ea2",
"_SELINUX_CONTEXT" : "unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023",
"_GID" : "1000",
"_TRANSPORT" : "syslog",
"_EXE" : "/usr/bin/sudo",
"MESSAGE" : "admin_trake : PWD=/home/admin_trake ; USER=root ; COMMAND=/bin/bash -c 'date; timedatectl
| grep synchron; firewall-cmd --get-default-zone; firewall-cmd --get-active-zones; firewall-cmd --zone=work -
-list-rich-rules; cat /etc/ssh/sshd_config.d/01-permitrootlogin.conf; grep -v ^# /etc/ssh/sshd_config.d/50-red
hat.conf | grep .; nmcli -g ipv4.method con show bond0'",
"_SYSTEMD_OWNER_UID" : "1000",
"_AUDIT_LOGINUID" : "1000",
"_BOOT_ID" : "006930a7d0734bf089fe2ab03b8fdaa2",
"_PID" : "1976",
"_SYSTEMD_SESSION" : "1",
"__REALTIME_TIMESTAMP" : "1793813194389265",
"_HOSTNAME" : "srv-core",
"_SYSTEMD_INVOCATION_ID" : "5d791b8c600847caaaea547fa7721138",
"_COMM" : "sudo",
"__MONOTONIC_TIMESTAMP" : "31937950",
"PRIORITY" : "5",
"_SYSTEMD_USER_SLICE" : "-.slice"
}
{
"_UID" : "1000",
"__MONOTONIC_TIMESTAMP" : "31940092",
"_SYSTEMD_USER_SLICE" : "-.slice",
"_AUDIT_SESSION" : "1",
"MESSAGE" : "pam_unix(sudo:session): session opened for user root(uid=0) by admin_trake(uid=1000)",
"_HOSTNAME" : "srv-core",
[root@srv-core ~]#

```
[root@srv-core ~]# journalctl -b -1 -p err --no-pager | tail -12
nov. 04 18:34:51 srv-core sshd[3325]: error: Bind to port 2242 on 192.168.56.101 failed: Permission denied.
nov. 04 18:34:51 srv-core sshd[3325]: fatal: Cannot bind any address.
nov. 04 18:34:51 srv-core systemd[1]: Failed to start OpenSSH server daemon.
nov. 04 18:35:33 srv-core sshd[3358]: error: Bind to port 2242 on 10.42.0.10 failed: Permission denied.
nov. 04 18:35:33 srv-core sshd[3358]: error: Bind to port 2242 on 192.168.56.101 failed: Permission denied.
nov. 04 18:35:33 srv-core sshd[3358]: fatal: Cannot bind any address.
nov. 04 18:35:33 srv-core systemd[1]: Failed to start OpenSSH server daemon.
nov. 04 18:36:15 srv-core sshd[3405]: error: Bind to port 2242 on 10.42.0.10 failed: Permission denied.
nov. 04 18:36:15 srv-core sshd[3405]: error: Bind to port 2242 on 192.168.56.101 failed: Permission denied.
nov. 04 18:36:15 srv-core sshd[3405]: fatal: Cannot bind any address.
nov. 04 18:36:15 srv-core systemd[1]: Failed to start OpenSSH server daemon.
nov. 04 18:50:50 srv-core dbus-broker-launch[736]: Activation request for 'org.freedesktop.nm_dispatcher' failed.
[root@srv-core ~]# lastb -n 15 ; last -F -n 10
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:21 - 13:21  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia    pts/1                        Tue Nov  3 13:14 - 13:14  (00:00)
nadia                                 Tue Nov  3 12:24 - 12:24  (00:00)
nadia                                 Tue Nov  3 12:24 - 12:24  (00:00)
nadia                                 Tue Nov  3 12:24 - 12:24  (00:00)

btmp commence Mon Nov  2 14:29:34 2026
reboot   system boot  5.14.0-687.10.1. Wed Nov  4 18:50:59 2026   still running
admin_tr tty1                         Wed Nov  4 18:33:52 2026 - down                      (00:16)
reboot   system boot  5.14.0-687.10.1. Wed Nov  4 18:26:06 2026 - Wed Nov  4 18:50:50 2026  (00:24)
admin_tr pts/1        192.168.56.1    Tue Nov  3 16:49:46 2026 - Tue Nov  3 16:50:00 2026  (00:00)
admin_tr tty1                         Tue Nov  3 16:30:21 2026 - down                    (1+01:54)
admin_tr pts/2        192.168.56.1    Tue Nov  3 13:19:56 2026 - Tue Nov  3 13:20:01 2026  (00:00)
admin_tr pts/0        192.168.56.1    Tue Nov  3 12:05:54 2026 - Tue Nov  3 16:23:22 2026  (04:17)
admin_tr pts/0        192.168.56.1    Mon Nov  2 20:40:31 2026 - Tue Nov  3 01:16:08 2026  (04:35)
reboot   system boot  5.14.0-687.10.1. Mon Nov  2 20:40:09 2026 - Wed Nov  4 18:24:58 2026 (1+21:44)
admin_tr pts/2        192.168.56.1    Mon Nov  2 20:21:00 2026 - Mon Nov  2 20:27:10 2026  (00:06)

wtmp commence Mon Nov  2 14:29:07 2026
[root@srv-core ~]# lslogins -u --output=USER,PWD-CHANGE,PWD-EXPIR,LAST-LOGIN,FAILED-LOGIN
USER          PWD-CHANGE    PWD-EXPIR   LAST-LOGIN FAILED-LOGIN
root         1969-déc.31
admin_trake  1969-déc.31                     18:33 nov.03/12:05
julien      nov.02/01:00              nov.03/15:16 nov.03/12:18
nadia       nov.03/01:00              nov.03/15:59 nov.03/13:21
tarek       nov.03/01:00              nov.03/14:59
audit-ext   nov.03/01:00 nov.24/01:00 nov.03/14:55
sysops            01:00                           nov.02/18:11
[root@srv-core ~]#
```

auditd

```
[root@srv-core ~]# systemctl is-enabled auditd ; systemctl is-active auditd ; ls -l /etc/audit/rules.d/
enabled
active
total 8
-rw-r-----. 1 root root 733  4 nov.  18:54 90-helios.rules
-rw-------. 1 root root 244  2 nov.  14:22 audit.rules
[root@srv-core ~]# auditctl -l ; auditctl -l | wc -l
-w /etc/passwd -p wa -k identity
-w /etc/shadow -p wa -k identity
-w /etc/group -p wa -k identity
-w /etc/gshadow -p wa -k identity
-w /etc/sudoers -p wa -k sudoers
-w /etc/sudoers.d -p wa -k sudoers
-w /etc/ssh/sshd_config -p wa -k sshd-config
-w /etc/ssh/sshd_config.d -p wa -k sshd-config
-w /usr/local/bin/helios-lastlog -p x -k helios-setgid
-a always,exit -F arch=b64 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F key=privesc
-a always,exit -F arch=b32 -S execve -F euid=0 -F auid>=1000 -F auid!=-1 -F key=privesc
-w /etc/firewalld -p wa -k firewall
12
[root@srv-core ~]# auditctl -s | head -5
enabled 1
failure 1
pid 708
rate_limit 0
backlog_limit 8192
[root@srv-core ~]#
```

provoquer des événements :

29

```
type=CONFIG_CHANGE msg=audit(04/11/2026 18:54:03.935:311) : auid=admin_trake ses=11 subj=unconfined_u:unconfined_r:unconfined_t:s0
-s0:c0.c1023 op=add_rule key=sudoers list=exit res=yes
----
type=PROCTITLE msg=audit(04/11/2026 18:54:03.936:312) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=PATH msg=audit(04/11/2026 18:54:03.936:312) : item=0 name=/etc/sudoers.d inode=876527 dev=fd:00 mode=dir,750 ouid=root ogid=r
oot rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:54:03.936:312) : cwd=/home/admin_trake
type=SOCKADDR msg=audit(04/11/2026 18:54:03.936:312) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(04/11/2026 18:54:03.936:312) : arch=x86_64 syscall=sendto success=yes exit=1080 a0=0x3 a1=0x7ffc9dee4e60 a2
=0x438 a3=0x0 items=1 ppid=2409 pid=2429 auid=admin_trake uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsg
id=root tty=(none) ses=11 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(nul
l)
type=CONFIG_CHANGE msg=audit(04/11/2026 18:54:03.936:312) : auid=admin_trake ses=11 subj=unconfined_u:unconfined_r:unconfined_t:s0
-s0:c0.c1023 op=add_rule key=sudoers list=exit res=yes
----
type=PROCTITLE msg=audit(04/11/2026 18:54:22.896:366) : proctitle=touch /etc/sudoers.d/test-audit
type=PATH msg=audit(04/11/2026 18:54:22.896:366) : item=1 name=/etc/sudoers.d/test-audit inode=876697 dev=fd:00 mode=file,640 ouid
=root ogid=root rdev=00:00 obj=unconfined_u:object_r:etc_t:s0 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_froo
tid=0
type=PATH msg=audit(04/11/2026 18:54:22.896:366) : item=0 name=/etc/sudoers.d/ inode=876527 dev=fd:00 mode=dir,750 ouid=root ogid=
root rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:54:22.896:366) : cwd=/home/admin_trake
type=SYSCALL msg=audit(04/11/2026 18:54:22.896:366) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x7ffe89447e97
a2=O_WRONLY|O_CREAT|O_NOCTTY|O_NONBLOCK a3=0x1b6 items=2 ppid=2502 pid=2503 auid=admin_trake uid=root gid=root euid=root suid=root
 fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=13 comm=touch exe=/usr/bin/touch subj=unconfined_u:unconfined_r:unconfin
ed_t:s0-s0:c0.c1023 key=sudoers
----
type=PROCTITLE msg=audit(04/11/2026 18:54:22.897:368) : proctitle=rm -f /etc/sudoers.d/test-audit
type=PATH msg=audit(04/11/2026 18:54:22.897:368) : item=1 name=/etc/sudoers.d/test-audit inode=876697 dev=fd:00 mode=file,640 ouid
=root ogid=root rdev=00:00 obj=unconfined_u:object_r:etc_t:s0 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_froo
tid=0
type=PATH msg=audit(04/11/2026 18:54:22.897:368) : item=0 name=/etc/sudoers.d/ inode=876527 dev=fd:00 mode=dir,750 ouid=root ogid=
root rdev=00:00 obj=system_u:object_r:etc_t:s0 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:54:22.897:368) : cwd=/home/admin_trake
type=SYSCALL msg=audit(04/11/2026 18:54:22.897:368) : arch=x86_64 syscall=unlinkat success=yes exit=0 a0=AT_FDCWD a1=0x5596ca4cc4d
0 a2=0x0 a3=0x100 items=2 ppid=2502 pid=2504 auid=admin_trake uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root
 fsgid=root tty=(none) ses=13 comm=rm exe=/usr/bin/rm subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=sudoers
[root@srv-core ~]# ausearch -k helios-setgid -i | tail -10
type=SOCKADDR msg=audit(04/11/2026 18:54:03.939:315) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(04/11/2026 18:54:03.939:315) : arch=x86_64 syscall=sendto success=yes exit=1100 a0=0x3 a1=0x7ffc9dee4e60 a2
=0x44c a3=0x0 items=1 ppid=2409 pid=2429 auid=admin_trake uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsg
id=root tty=(none) ses=11 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=(nul
l)
type=CONFIG_CHANGE msg=audit(04/11/2026 18:54:03.939:315) : auid=admin_trake ses=11 subj=unconfined_u:unconfined_r:unconfined_t:s0
-s0:c0.c1023 op=add_rule key=helios-setgid list=exit res=yes
----
type=PROCTITLE msg=audit(04/11/2026 18:54:22.975:377) : proctitle=/usr/local/bin/helios-lastlog
type=PATH msg=audit(04/11/2026 18:54:22.975:377) : item=1 name=/lib64/ld-linux-x86-64.so.2 inode=50597478 dev=fd:00 mode=file,755
ouid=root ogid=root rdev=00:00 obj=system_u:object_r:ld_so_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_fr
ootid=0
type=PATH msg=audit(04/11/2026 18:54:22.975:377) : item=0 name=/usr/local/bin/helios-lastlog inode=1043294 dev=fd:00 mode=file,sgi
d,755 ouid=root ogid=telemetry rdev=00:00 obj=unconfined_u:object_r:bin_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_
fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:54:22.975:377) : cwd=/home/julien
type=EXECVE msg=audit(04/11/2026 18:54:22.975:377) : argc=1 a0=/usr/local/bin/helios-lastlog
type=SYSCALL msg=audit(04/11/2026 18:54:22.975:377) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x5606de176d70 a1=0x5606de1
78280 a2=0x5606de17e6c0 a3=0x1b6 items=2 ppid=2506 pid=2535 auid=admin_trake uid=julien gid=transport euid=julien suid=julien fsui
d=julien egid=telemetry sgid=telemetry fsgid=telemetry tty=(none) ses=13 comm=helios-lastlog exe=/usr/local/bin/helios-lastlog sub
j=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 key=helios-setgid
[root@srv-core ~]#
```

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```
[root@srv-core ~]# ausearch -k privesc -i --start today | tail -14
----
type=PROCTITLE msg=audit(04/11/2026 18:55:36.920:539) : proctitle=ausearch -k privesc -i --start today
type=PATH msg=audit(04/11/2026 18:55:36.920:539) : item=1 name=/lib64/ld-linux-x86-64.so.2 inode=50597478 dev=fd:00 mode=file,755
ouid=root ogid=root rdev=00:00 obj=system_u:object_r:ld_so_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_fr
ootid=0
type=PATH msg=audit(04/11/2026 18:55:36.920:539) : item=0 name=/sbin/ausearch inode=50965124 dev=fd:00 mode=file,755 ouid=root ogi
d=root rdev=00:00 obj=system_u:object_r:bin_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:55:36.920:539) : cwd=/home/admin_trake
type=EXECVE msg=audit(04/11/2026 18:55:36.920:539) : argc=6 a0=ausearch a1=-k a2=privesc a3=-i a4=--start a5=today
type=SYSCALL msg=audit(04/11/2026 18:55:36.920:539) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x564c40c092a0 a1=0x564c40c
05e10 a2=0x564c40c08b70 a3=0x8 items=2 ppid=2779 pid=2780 auid=admin_trake uid=root gid=root euid=root suid=root fsuid=root egid=r
oot sgid=root fsgid=root tty=(none) ses=19 comm=ausearch exe=/usr/sbin/ausearch subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:
c0.c1023 key=privesc
----
type=PROCTITLE msg=audit(04/11/2026 18:55:36.920:540) : proctitle=tail -14
type=PATH msg=audit(04/11/2026 18:55:36.920:540) : item=1 name=/lib64/ld-linux-x86-64.so.2 inode=50597478 dev=fd:00 mode=file,755
ouid=root ogid=root rdev=00:00 obj=system_u:object_r:ld_so_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_fr
ootid=0
type=PATH msg=audit(04/11/2026 18:55:36.920:540) : item=0 name=/bin/tail inode=34331482 dev=fd:00 mode=file,755 ouid=root ogid=roo
t rdev=00:00 obj=system_u:object_r:bin_t:s0 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(04/11/2026 18:55:36.920:540) : cwd=/home/admin_trake
type=EXECVE msg=audit(04/11/2026 18:55:36.920:540) : argc=2 a0=tail a1=-14
type=SYSCALL msg=audit(04/11/2026 18:55:36.920:540) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x564c40c090b0 a1=0x564c40c
09960 a2=0x564c40c08b70 a3=0x8 items=2 ppid=2779 pid=2781 auid=admin_trake uid=root gid=root euid=root suid=root fsuid=root egid=r
oot sgid=root fsgid=root tty=(none) ses=19 comm=tail exe=/usr/bin/tail subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
key=privesc
[root@srv-core ~]#
```

aureport

31

```
[root@srv-core ~]# ausearch -m USER_LOGIN,USER_AUTH --start today -i | tail -6
----
type=USER_LOGIN msg=audit(04/11/2026 18:55:36.888:530) : pid=2737 uid=root auid=admin_trake ses=19 subj=system_u:system_r:sshd_t:s
0-s0:c0.c1023 msg='op=login id=admin_trake exe=/usr/libexec/openssh/sshd-session hostname=? addr=192.168.56.1 terminal=ssh res=suc
cess'
----
type=USER_AUTH msg=audit(04/11/2026 18:55:53.692:555) : pid=2799 uid=root auid=unset ses=unset subj=system_u:system_r:sshd_t:s0-s0
:c0.c1023 msg='op=pubkey_auth grantors=auth-key acct=admin_trake exe=/usr/libexec/openssh/sshd-session hostname=? addr=192.168.56.
1 terminal=? res=success'
----
type=USER_LOGIN msg=audit(04/11/2026 18:55:53.878:574) : pid=2799 uid=root auid=admin_trake ses=21 subj=system_u:system_r:sshd_t:s
0-s0:c0.c1023 msg='op=login id=admin_trake exe=/usr/libexec/openssh/sshd-session hostname=? addr=192.168.56.1 terminal=ssh res=suc
cess'
[root@srv-core ~]# aureport --failed --summary -i | head -12 ; aureport -k --summary -i

Failed Summary Report
======================
Range of time in logs: 02/11/2026 14:29:09.926 - 04/11/2026 18:55:53.919
Selected time for report: 02/11/2026 14:29:09 - 04/11/2026 18:55:53.919
Number of changes in configuration: 0
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 25
Number of authentications: 0
Number of failed authentications: 38
Number of users: 2

Key Summary Report
===========================
total  key
===========================
52  privesc
4  identity
4  sudoers
2  sshd-config
2  helios-setgid
1  firewall
[root@srv-core ~]#
```

script + timer

32

```
[root@srv-core ~]# install -m 750 -o root -g root /tmp/m3/helios-watch-auth.sh /usr/local/sbin/helios-watch-auth.sh
[root@srv-core ~]# install -m 644 -o root -g root /tmp/m3/helios-watch-auth.service /tmp/m3/helios-watch-auth.timer /etc
/systemd/system/
[root@srv-core ~]# bash -n /usr/local/sbin/helios-watch-auth.sh && echo "bash -n OK"
bash -n OK
[root@srv-core ~]# systemd-analyze verify /etc/systemd/system/helios-watch-auth.service /etc/systemd/system/helios-watch
-auth.timer && echo "verify OK"
verify OK
[root@srv-core ~]# systemctl daemon-reload ; systemctl enable --now helios-watch-auth.timer
Created symlink /etc/systemd/system/timers.target.wants/helios-watch-auth.timer → /etc/systemd/system/helios-watch-auth.
timer.
[root@srv-core ~]# systemctl list-timers helios-watch-auth.timer --no-pager
NEXT                        LEFT          LAST PASSED UNIT                    ACTIVATES
Wed 2026-11-04 18:58:00 CET 1min 36s left -    -      helios-watch-auth.timer helios-watch-auth.service

1 timers listed.
Pass --all to see loaded but inactive timers, too.
[root@srv-core ~]#
```

codes retour

33

```
[root@srv-core ~]# /usr/local/sbin/helios-watch-auth.sh ; echo "rc nominal = $?"
rc nominal = 0
[root@srv-core ~]# flock /run/lock/helios-watch-auth.lock sleep 5 & sleep 1 ; /usr/local/sbin/helios-watch-auth.sh ; ech
o "rc deja en cours = $?" ; wait
rc deja en cours = 2
[root@srv-core ~]# journalctl -t helios-watch-auth -n 5 --no-pager
nov. 04 18:56:40 srv-core helios-watch-auth[3070]: analyse terminee : 0 nouvelle(s) adresse(s) bloquee(s)
nov. 04 18:56:41 srv-core helios-watch-auth[3076]: une autre instance tourne deja
[root@srv-core ~]#
```

attaque depuis srv-edge

34

```
[sysops@srv-edge ~]$ for i in $(seq 1 10); do ssh -p 2242 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout
=3 pirate@192.168.56.101 true 2>/dev/null ; echo "tentative $i : rc=$?" ; done
tentative 1 : rc=255
tentative 2 : rc=255
tentative 3 : rc=255
tentative 4 : rc=255
tentative 5 : rc=255
tentative 6 : rc=255
tentative 7 : rc=255
tentative 8 : rc=255
tentative 9 : rc=255
tentative 10 : rc=255
[sysops@srv-edge ~]$
```

2e vague

35

```
[sysops@srv-edge ~]$ for i in $(seq 1 10); do ssh -p 2242 -o BatchMode=yes -o StrictHostKeyChecking=no -o ConnectTimeout
=3 pirate@192.168.56.101 true 2>/dev/null ; echo "tentative $i : rc=$?" ; done
tentative 1 : rc=255
tentative 2 : rc=255
tentative 3 : rc=255
tentative 4 : rc=255
tentative 5 : rc=255
tentative 6 : rc=255
tentative 7 : rc=255
tentative 8 : rc=255
tentative 9 : rc=255
tentative 10 : rc=255
[sysops@srv-edge ~]$
```

srv-edge bloquée

36

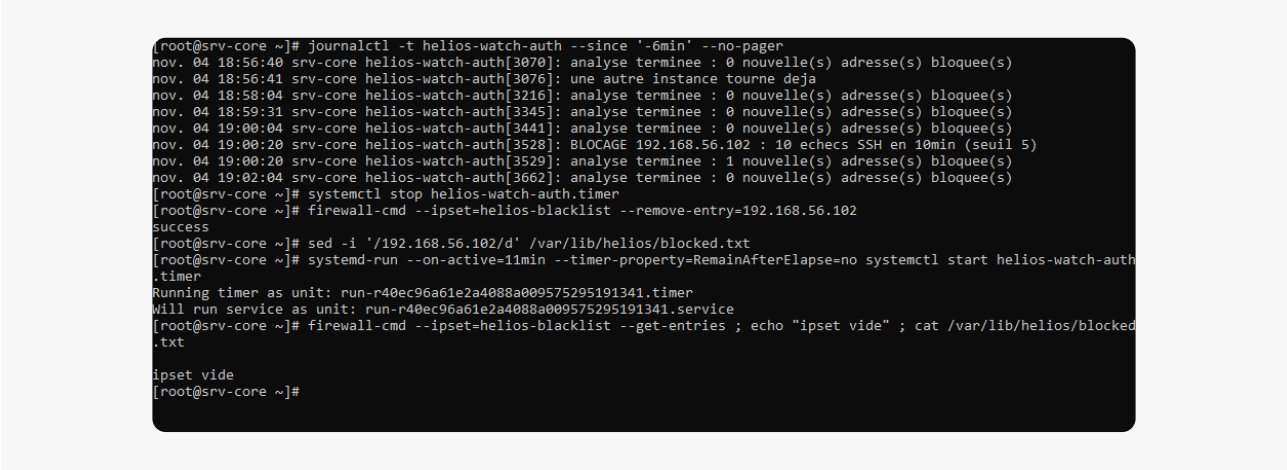
```
[root@srv-core ~]# journalctl -u sshd --since '-5min' -g 'Invalid user' --no-pager | tail
nov. 04 18:57:37 srv-core sshd-session[3124]: Invalid user pirate from 192.168.56.102 port 56918
nov. 04 18:57:37 srv-core sshd-session[3126]: Invalid user pirate from 192.168.56.102 port 56926
nov. 04 18:57:37 srv-core sshd-session[3128]: Invalid user pirate from 192.168.56.102 port 56930
nov. 04 18:57:37 srv-core sshd-session[3130]: Invalid user pirate from 192.168.56.102 port 56944
nov. 04 18:57:37 srv-core sshd-session[3132]: Invalid user pirate from 192.168.56.102 port 56950
nov. 04 19:00:05 srv-core sshd-session[3442]: Invalid user pirate from 192.168.56.102 port 53850
nov. 04 19:00:05 srv-core sshd-session[3444]: Invalid user pirate from 192.168.56.102 port 53866
nov. 04 19:00:05 srv-core sshd-session[3446]: Invalid user pirate from 192.168.56.102 port 53882
nov. 04 19:00:05 srv-core sshd-session[3449]: Invalid user pirate from 192.168.56.102 port 53884
nov. 04 19:00:06 srv-core sshd-session[3451]: Invalid user pirate from 192.168.56.102 port 53896
[root@srv-core ~]# systemctl start helios-watch-auth.service
[root@srv-core ~]# journalctl -t helios-watch-auth -n 10 --no-pager
nov. 04 18:56:40 srv-core helios-watch-auth[3070]: analyse terminee : 0 nouvelle(s) adresse(s) bloquee(s)
nov. 04 18:56:41 srv-core helios-watch-auth[3076]: une autre instance tourne deja
nov. 04 18:58:04 srv-core helios-watch-auth[3216]: analyse terminee : 0 nouvelle(s) adresse(s) bloquee(s)
nov. 04 18:59:31 srv-core helios-watch-auth[3345]: analyse terminee : 0 nouvelle(s) adresse(s) bloquee(s)
nov. 04 19:00:04 srv-core helios-watch-auth[3441]: analyse terminee : 0 nouvelle(s) adresse(s) bloquee(s)
nov. 04 19:00:20 srv-core helios-watch-auth[3528]: BLOCAGE 192.168.56.102 : 10 echecs SSH en 10min (seuil 5)
nov. 04 19:00:20 srv-core helios-watch-auth[3529]: analyse terminee : 1 nouvelle(s) adresse(s) bloquee(s)
[root@srv-core ~]# firewall-cmd --ipset=helios-blacklist --get-entries ; cat /var/lib/helios/blocked.txt
192.168.56.102
192.168.56.102
[root@srv-core ~]#
```

37

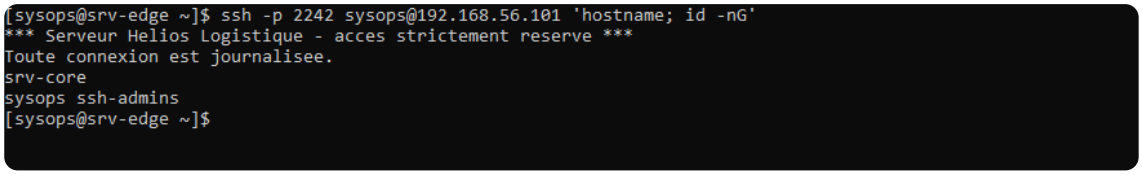
```
[sysops@srv-edge ~]$ nc -zv -w3 192.168.56.101 2242 ; echo "rc=$?"
Ncat: Version 7.92 ( https://nmap.org/ncat )
Ncat: TIMEOUT.
rc=1
[sysops@srv-edge ~]$ ssh -p 2242 -o ConnectTimeout=5 sysops@192.168.56.101 hostname ; echo "rc=$?"
ssh: connect to host 192.168.56.101 port 2242: Connection timed out
rc=255
[sysops@srv-edge ~]$
```

débloquer :

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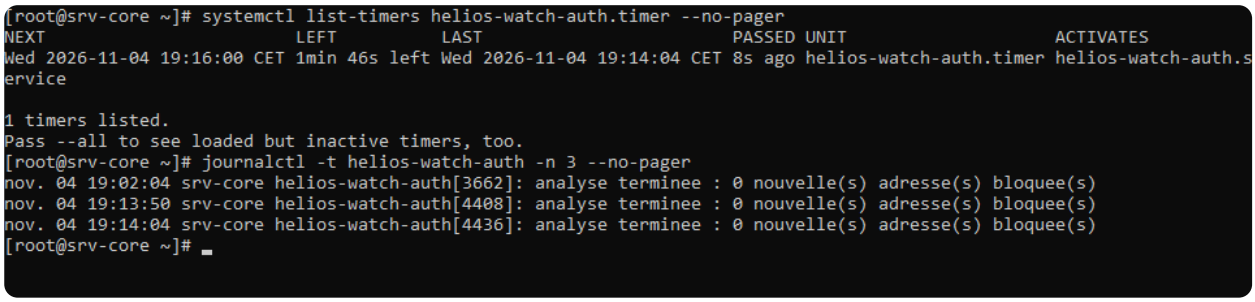


39



timer relancé

40



vérif finale

41

```
[root@srv-core ~]# firewall-cmd --get-default-zone ; firewall-cmd --get-active-zones
drop
drop
  interfaces: enp0s3 enp0s10 enp0s9
internal
  interfaces: bond0 bond0.42
work
  interfaces: enp0s8
  sources: 192.168.56.0/24
[root@srv-core ~]# sshd -T | grep -E '^(port|passwordauthentication|permitrootlogin|allowgroups) '
port 2242
permitrootlogin no
passwordauthentication no
allowgroups ssh-admins
[root@srv-core ~]# semanage port -l | grep ssh_port_t
ssh_port_t                     tcp      2242, 22
[root@srv-core ~]# journalctl -k -g SSH-NEW --no-pager | grep -c SSH-NEW
33
[root@srv-core ~]# journalctl --list-boots --no-pager
IDX BOOT ID                          FIRST ENTRY                 LAST ENTRY
 -1 006930a7d0734bf089fe2ab03b8fdaa2 Wed 2026-11-04 18:26:04 CET Wed 2026-11-04 18:50:50 CET
  0 bab8b16e80f4488c889c0027fcbc118d Wed 2026-11-04 18:50:56 CET Wed 2026-11-04 19:14:30 CET
[root@srv-core ~]# auditctl -l | wc -l ; ausearch -k sudoers -i | grep -c 'type=SYSCALL'
12
4
[root@srv-core ~]# systemctl list-timers helios-watch-auth.timer --no-pager | head -2
NEXT                         LEFT          LAST                         PASSED  UNIT                        ACTIVATES
Wed 2026-11-04 19:16:00 CET 1min 29s left Wed 2026-11-04 19:14:04 CET 25s ago helios-watch-auth.timer helios-watch-auth.
service
[root@srv-core ~]#
```

nettoyage sudo

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```
[root@srv-core ~]# rm -rf /tmp/10-helios-hardening.conf /tmp/05-helios-crypto.conf /tmp/issue.net /tmp/50-heli
os.conf /tmp/90-helios.rules /tmp/m3 /tmp/zones.txt
[root@srv-core ~]# rm -f /etc/sudoers.d/00-tp3-temp ; visudo -c ; ls -l /etc/sudoers.d/
Attention : /etc/sudoers.d/90-helios-ops:10:12: Cmnd_Alias « DANGEREUX » n'est pas utilisé
/etc/sudoers : analyse réussie
/etc/sudoers.d/90-helios-ops : analyse réussie
/etc/sudoers.d/91-nadia-edition : analyse réussie
total 8
-r--r-----. 1 root root 1139  3 nov.  15:09 90-helios-ops
-r--r-----. 1 root root   92  3 nov.  15:54 91-nadia-edition
[root@srv-core ~]#
```